JAMA | Original Investigation

Intravenous Tirofiban After Tenecteplase in Acute Ischemic Stroke
The INSTANT Randomized Clinical Trial

INSTANT Trial Authors for the INSTANT Investigators

IMPORTANCE Although recent trials have shown benefit with early tirofiban after thrombolysis, its efficacy remains uncertain in patients with acute ischemic stroke who do not have a large or medium vessel occlusion or a cardioembolic source and who show an inadequate clinical response to intravenous tenecteplase.

OBJECTIVE To assess the efficacy and safety of intravenous tirofiban administered after an inadequate response to intravenous tenecteplase in this specific patient population.

DESIGN, SETTING, AND PARTICIPANTS Investigator-initiated, randomized, double-blind, placebo-controlled trial conducted at 37 hospitals in China, enrolling 359 patients with acute ischemic stroke, without large or medium vessel occlusion or cardioembolic etiology, and with an insufficient clinical response to intravenous tenecteplase (defined as no significant change from baseline, neurological deterioration, or neurological fluctuation based on serial assessment of the National Institutes of Health Stroke Scale score within 4-24 hours after infusion). Recruitment took place between April 24, 2024, and July 16, 2025, with final follow-up on October 11, 2025.

INTERVENTIONS Intravenous tirofiban (n = 177), administered as a 0.3-μg/kg/min bolus over 30 minutes, followed by a continuous infusion of 0.075 μg/kg/min for up to 47.5 hours, or matching placebo (n = 182), initiated within 4 to 24 hours after intravenous tenecteplase. Oral antiplatelet therapy (aspirin and/or clopidogrel) was started 24 hours after thrombolysis in the placebo group and 44 hours after thrombolysis in the tirofiban group and continued in all patients through 90-day follow-up.

MAIN OUTCOMES AND MEASURES The primary outcome was an excellent outcome (defined as a score of 0 or 1 on the modified Rankin Scale; range, 0-6, with higher scores indicating more severe disability) at 90 days. Safety outcomes included incidence of symptomatic intracranial hemorrhage within 48 hours and 90-day mortality.

RESULTS Among 359 patients randomized (mean age, 66 years; 141 females [39.3%]), 358 (99.7%) completed the trial. An excellent outcome at 90 days was observed in 113 patients (63.8%) in the tirofiban group and in 95 patients (52.2%) in the placebo group (risk ratio, 1.22; 95% CI, 1.02-1.46; *P* = .03). Symptomatic intracranial hemorrhage within 48 hours occurred in 1 patient (0.9%) in the tirofiban group and no patients in the placebo group; 90-day mortality was 0.6% and 1.6%, respectively.

CONCLUSIONS AND RELEVANCE Among patients with acute ischemic stroke without large or medium vessel occlusion or a cardioembolic source who had an inadequate clinical response to intravenous tenecteplase, adjunctive intravenous tirofiban increased the likelihood of an excellent outcome at 90 days.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: NCT05604638

Visual Abstract
Research Summary
Supplemental content

Group Information: The INSTANT Trial Authors are listed at the end of this article. The INSTANT Investigators are listed in Supplement 3.

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Intravenous thrombolysis is the first-line treatment for acute ischemic stroke, with tenecteplase emerging as an effective alternative to alteplase based on recent large-scale randomized clinical trials and current international guidelines, including those from the European Stroke Organisation.[1-3] Nevertheless, a considerable proportion of patients show an insufficient clinical response within 24 hours after intravenous thrombolysis, presenting as no significant improvement, neurological deterioration, or fluctuation, which is associated with unfavorable functional outcomes.[4] This limited response may be attributable to mechanisms such as recanalization failure, reocclusion, or microcirculatory impairment due to residual fibrinogen and platelet aggregation.[5,6] In this context, early antiplatelet therapy may be an effective intervention to counteract these pathological mechanisms and potentially improve outcomes.

Tirofiban, a rapidly acting glycoprotein IIb/IIIa inhibitor with a short half-life, offers a promising pharmacological profile for early intervention. Although widely used in myocardial infarction, its application in acute ischemic stroke is off label. The Safety of Tirofiban in Acute Ischemic Stroke trial showed an acceptable safety profile for tirofiban as a standalone therapy in stroke,[7] and several studies indicated its potential benefit in improving functional outcomes in select patients.[8,9] The ASSET-IT trial demonstrated that for patients with acute ischemic noncardioembolic stroke who are not eligible for thrombectomy, intravenous tirofiban within 60 minutes after thrombolytic therapy (with either alteplase or tenecteplase) yielded favorable functional outcomes at 90 days. However, the existing evidence is limited by nonstandard thrombolytic dosing and limited sample size, leaving the combination of intravenous thrombolysis and tirofiban inadequately investigated.[10]

Thus, we designed and conducted the Intravenous Tenecteplase and Tirofiban for Acute Ischemic Stroke (INSTANT) trial to evaluate the efficacy and safety of early tirofiban administration following intravenous tenecteplase in patients with acute ischemic stroke with a limited response to tenecteplase. To focus on a pathophysiologically relevant subgroup, we selected patients without large or medium vessel occlusion or a cardioembolic source. This design focused the study on a population in whom underlying mechanisms such as microcirculatory impairment might be most responsive to adjunctive antiplatelet therapy with tirofiban.

Key Points

Question Among patients with acute ischemic stroke without large or medium vessel occlusion or cardioembolic source, does intravenous tirofiban after an insufficient response to intravenous tenecteplase improve functional outcomes?

Findings In this randomized trial of 359 patients, tirofiban significantly increased the proportion of patients with an excellent outcome (defined as a score of 0 or 1 on the modified Rankin Scale) at 90 days compared with placebo (63.8% vs 52.2%, respectively).

Meaning These findings suggest a potential benefit of adjunctive intravenous tirofiban after intravenous tenecteplase in this specific patient population and warrant further investigation.

Methods

Trial Design and Oversight

INSTANT was a multicenter randomized clinical trial with a double-blind, double-dummy design. The trial was conducted across 37 hospitals in China. The trial protocol (Supplement 1) received approval from the medical ethics committees at Ganzhou People's Hospital and each participating site.[11] A protocol amendment was made in May 2025. This amendment, approved by the steering committee and ethics committees before any outcome data analysis, refined the eligibility criteria pertaining to the definition of an inadequate clinical response after intravenous tenecteplase. Written informed consent was obtained from all enrolled patients or their legally authorized representatives prior to randomization. The trial was conducted in compliance with the Declaration of Helsinki[12] and International Council for Harmonisation Good Clinical Practice guidelines and reported in accordance with the Consolidated Standards of Reporting Trials (CONSORT) reporting guideline.[13] The statistical analysis plan is available in Supplement 2.

An independent data and safety monitoring board oversaw trial progress, evaluated adverse events, and ensured the study's integrity. Efficacy and safety outcomes, procedure-related complications, and serious adverse events were adjudicated by an independent clinical events committee blinded to group allocation. All neuroimaging data were assessed by an independent core laboratory that was blinded to group assignments. Tirofiban and its matching placebo (saline), along with aspirin (Bayer), clopidogrel (Sanofi), and their respective placebos, were all supplied in identically appearing packaging (including labeling, dosage, size, weight, and color) and differentiated only by unique identification numbers. Tirofiban, saline placebo, and the oral antiplatelet placebos were manufactured and provided by Lunan Pharmaceutical Group Co Ltd, while active aspirin and clopidogrel were sourced from their originator manufacturers. Lists of the trial sites, investigators, and administrative staff are provided in eAppendix 1 in Supplement 3.

Patients

Demographic, age, and sex data were collected. All patients received intravenous tenecteplase as a bolus over 5 to 10 seconds at a dose of 0.25 mg/kg (maximum dose, 25 mg). The protocol mandated a minimum interval of 4 hours between tenecteplase completion and initiation of the study drug (tirofiban or placebo). Eligible participants were aged 18 years or older with acute ischemic stroke; had a National Institutes of Health Stroke Scale (NIHSS; score range, 0-42, with higher scores denoting more severe neurological deficits) score of 4 or higher before randomization; demonstrated an insufficient clinical response to intravenous tenecteplase, defined as no significant change from baseline (an increase or decrease of 0 or 1 point in the NIHSS score), neurological deterioration

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(NIHSS worsened by ≥2 points), or neurological fluctuation (a ≥4-point increase in the NIHSS score followed by a ≥4-point decrease or a ≥4-point decrease followed by a ≥4-point increase); and were assessed by trained site investigators within 4 to 24 hours after infusion. Centralized adjudication ensured consistency across sites in determining final eligibility based on these criteria.

Key exclusion criteria were intracranial hemorrhage after intravenous thrombolysis but before randomization, confirmed or suspected cardioembolic stroke, and, importantly, large or medium vessel occlusion (internal carotid artery, middle cerebral artery M1/M2/M3, anterior cerebral artery A1/A2/A3, posterior cerebral artery P1/P2/P3, vertebral or basilar artery) on computed tomography angiography/magnetic resonance angiography/digital subtraction angiography, thereby focusing the trial on a population in whom endovascular thrombectomy was neither indicated nor performed. Detailed selection criteria and the causes of stroke are provided in eAppendix 2 in Supplement 3.

Randomization and Masking

All eligible patients were randomly assigned to either the tirofiban group or placebo group in a 1:1 ratio. Randomization was performed using stratified block randomization (block size of 4), with stratification by participating site. The allocation sequence was concealed, and trial personnel, participants, and outcome assessors were blinded to treatment allocation. The study drug infusion was initiated as soon as possible after randomization (ideally within 10 minutes).

Treatments

Patients in the tirofiban group received an intravenous bolus of tirofiban at a dose of 0.3 µg/kg/min over 30 minutes, followed by a continuous infusion of 0.075 µg/kg/min for up to 47.5 hours. The placebo group received a matching saline placebo following the same intravenous bolus and infusion regimen. As part of the double-dummy design, oral antiplatelet therapy was administered as follows: 24 hours after intravenous tenecteplase, patients who initially received intravenous placebo were administered active antiplatelet therapy (aspirin at 100 mg, clopidogrel at 75 mg, or both) and patients who initially received tirofiban were administered matching placebos. Then, beginning 44 hours after randomization, all patients received active antiplatelet therapy (aspirin, 100 mg once daily; clopidogrel, 75 mg once daily; or both), which was continued through the 90-day follow-up. The decision to initiate either single or dual antiplatelet therapy was at the discretion of treating physicians, in accordance with clinical practice guidelines and patients' clinical characteristics, and was independent of study randomization. Use of other anticoagulants or antiplatelet agents was prohibited during the first 48 hours after randomization. eFigure 1 in Supplement 3 provides the treatment flow diagram.

Outcomes

The primary efficacy end point was an excellent outcome, defined as a modified Rankin Scale (mRS) score of 0 or 1 at 90 days. The mRS scores were centrally adjudicated by 2 certified neurologists using standardized video or voice recordings at 90-day follow-up.[14] Secondary efficacy end points at 90 days were the ordinal distribution of disability scores on the mRS (shift analysis), functional independence (mRS score of 0-2), independent ambulation and self-care (mRS score of 0-3), early neurological improvement (≥30% reduction in NIHSS score at 48 hours after randomization),[15] and health-related quality of life, assessed using the European Quality of Life 5-Dimension 5-Level (EQ-5D-5L; range, −0.39 to 1, with higher scores indicating better quality of life) scale at 90 days.

Safety outcomes were symptomatic intracranial hemorrhage within 48 hours (defined by the Heidelberg Bleeding Classification),[16] any intracranial hemorrhage within 48 hours, mortality at 90 days, and any serious or adverse event. Follow-up imaging was performed for suspected neurological deterioration or per clinical discretion. All safety events were reviewed by the independent blinded clinical events committee.

Sample Size Calculation

The sample size was estimated based on a previous observational study, assuming proportions of 38% and 23% for the primary efficacy end point (90-day mRS of 0-1) in the tirofiban and placebo groups, respectively.[17] To enhance the reliability and robustness of the trial findings, the statistical power was increased from 80% to 85%. With a 2-sided statistical significance level of $\alpha = .05$ and 85% power, and accounting for an attrition rate of 5%, a total of 348 patients (174 in each group) were required.

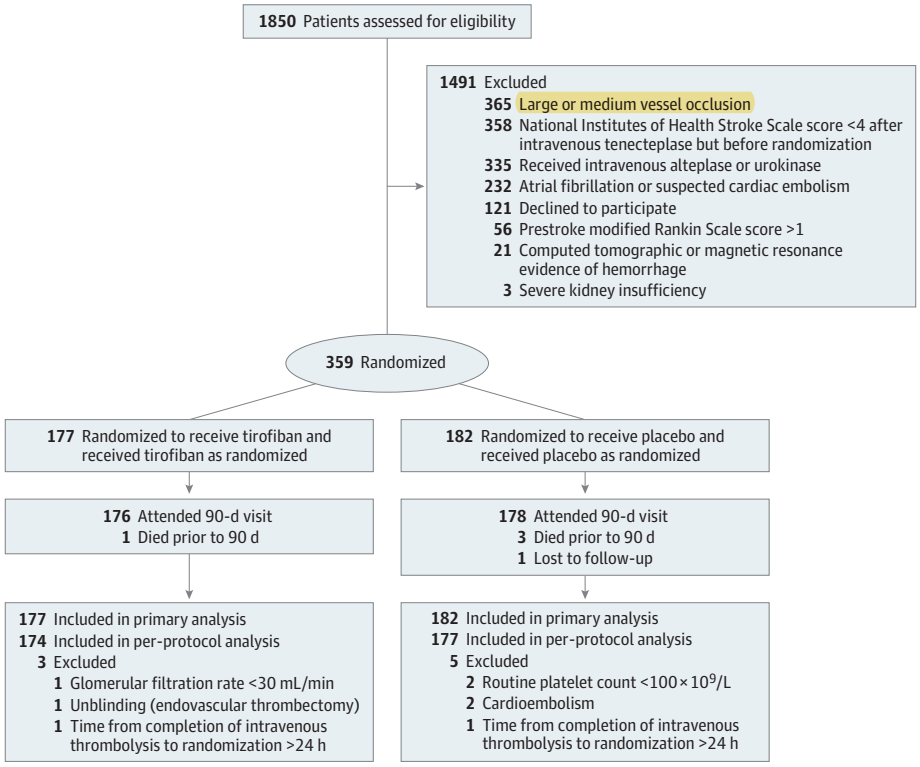
Statistical Analysis

The primary outcome and binary secondary outcomes were analyzed using a modified Poisson regression model to generate risk ratios as measurements of treatment effect.[18] As prespecified in the statistical analysis plan (Supplement 2), the primary end-point analysis was performed based on the unadjusted risk ratio in the full analysis set. This prespecified approach was chosen because it provides reliable and stable estimates of treatment effects while allowing for adjustment for prespecified covariates. The model is well validated for analyzing binary outcomes in randomized trials. Given that the number of intracranial hemorrhage events was zero in the placebo group, Firth penalized likelihood Poisson regression was used to mitigate parameter estimation bias arising from small sample sizes and zero events. For the ordinal analysis of the mRS score at 90 days, the generalized odds ratio was used. This metric represents the odds that a randomly selected patient in the tirofiban group achieves a lower mRS score than a randomly selected patient in the placebo group, accounting for all possible between-group pairwise comparisons; ties were resolved by assigning half a win to each group. A win odds greater than 1 indicates a favorable treatment effect associated with tirofiban. The EQ-5D-5L score at 90 days was evaluated using the win ratio method.[19] This analysis compared all possible patient pairs between the treatment groups, with the patient having the higher (more favorable) score deemed the winner in each pair. The win ratio is defined as the total wins for the tirofiban group

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Figure 1. Flow Diagram of Participants in the INSTANT Trial



INSTANT indicates Intravenous Tenecteplase and Tirofiban for Acute Ischemic Stroke.

divided by the total wins for the placebo group; detailed components (wins, losses, and ties) are provided in eTable 1 in Supplement 3. Both unadjusted and adjusted treatment effects and their 95% confidence intervals are reported, with adjustments made for the following prespecified covariates: age, baseline NIHSS score, and duration from last known well time to randomization. Adjusted analyses for the generalized odds ratio and win ratio used the inverse probability of treatment weighting method. This approach was used not to address potential confounding, which is adequately controlled through randomization, but as a robust statistical technique to incorporate the prespecified covariates into the analysis of these complex end points, thereby potentially increasing the precision of the treatment effect estimate. The number needed to treat for the mRS 0-1 outcome was derived by taking the inverse of the absolute risk reduction. Given the absence of multiplicity correction for the analyses of secondary outcomes, these results should be considered exploratory, with an understanding of the increased potential for type I error.

The primary analysis was based on the full analysis set, which included all randomized patients analyzed according to their assigned treatment group. One missing 90-day mRS score was imputed using the multiple imputation method to create a single dataset for the mRS analysis. Patients who received the randomized treatment and did not have major protocol violations were included in the per-protocol analysis. Prespecified subgroup analyses assessed treatment effect modification by age, baseline NIHSS score, duration from last known well time to randomization, and, following a protocol amendment in May 2025, duration from last known well time to administration of tirofiban or placebo and type of antiplatelet therapy after administration of tirofiban or placebo. Sensitivity analyses included complete-case, worst-case, and best-case scenarios. All tests were 2-sided with a significance level of $\alpha = .05$. Analyses were performed using SAS version 9.4 (SAS Institute Inc) and R version 4.1.1. (R Foundation). The statistical analysis code is available in eAppendix 2 in Supplement 3.

Results

Patients

From April 24, 2024, through July 16, 2025, a total of 1850 patients were screened for eligibility at 37 centers in China, of whom 1491 were excluded for various reasons (**Figure 1**; eFigure 2 in Supplement 3). A total of 359 patients underwent randomization (177 to the tirofiban group and 182 to the placebo group). The mean age of the patients was 66 (SD, 11) years and 141 (39.3%) were female. No crossover to the alternative treatment strategy or failure to administer the assigned study drug occurred. One patient in the placebo group was lost to 90-day follow-up.

Baseline Characteristics

The baseline characteristics were balanced between the 2 groups (**Table 1**). The representativeness of the randomized

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patients is summarized in eTable 2 in Supplement 3. The median NIHSS score at baseline was 6 (IQR, 5-9) in both groups. The median duration from last known well time to intravenous tenecteplase was 180 (IQR, 120-242) minutes in the tirofiban group and 180 (IQR, 122-227) minutes in the placebo group. The median duration from last known well time to randomization was 638 (IQR, 460-1051) minutes in the tirofiban group and 623 (IQR, 480-1001) minutes in the placebo group. The cause of stroke was small vessel occlusion in 73.3% of patients, large artery atherosclerosis in 11.0%, unknown in 15.0%, and cardioembolism in 0.6%.

Primary Outcome

The proportion of patients with excellent outcome (ie, mRS score of 0-1) at 90 days was 63.8% in the tirofiban group and 52.2% in the placebo group (risk ratio, 1.22; 95% CI, 1.02-1.46; P = .03) (**Table 2**). The number needed to treat for 1 additional patient to have an excellent outcome was 8.7 (95% CI, 4.6-72.1). In the prespecified adjusted analysis, the risk ratio was 1.17 (95% CI, 0.98-1.41; P = .08) (eTable 3 in Supplement 3). The per-protocol analysis yielded results similar to the primary analysis (eTable 4 in Supplement 3), and the effect estimates for tirofiban remained consistent across worst-case, best-case, and complete-case sensitivity analyses (eTable 5 in Supplement 3). The results of the prespecified subgroup analyses of excellent outcome at 90 days are shown in **Figure 2**.

Secondary Outcomes

Scores on the mRS at 90 days were a median of 1 (IQR, 1-2) and a mean of 1.4 (SD, 1.3) in the tirofiban group and a median of 1 (IQR, 1-3) and a mean of 1.7 (SD, 1.4) in the placebo group. The generalized odds ratio for a more favorable outcome was 1.38 (95% CI, 1.01-1.89) (Table 2 and **Figure 3**). Early neurological improvement occurred in 54.2% of patients in the tirofiban group and in 50.0% of those in the placebo group (risk ratio, 1.08; 95% CI, 0.89-1.32). Regarding 90-day health-related quality of life, assessed by the EQ-5D-5L scale, the tirofiban group had a win proportion of 54.7% compared with 38.5% in the placebo group (win ratio, 1.42; 95% CI, 1.08-1.83). The proportion of patients achieving functional independence (ie, an mRS score of 0-2) at 90 days was 78.5% in the tirofiban group and 72.0% in the placebo group (risk ratio, 1.09; 95% CI, 0.97-1.23; P = .14). The proportion achieving independent ambulation and self-care (ie, an mRS score of 0-3) at 90 days was 93.2% with tirofiban and 89.6% with placebo (risk ratio, 1.04; 95% CI, 0.98-1.11; P = .21).

Safety Outcomes

Symptomatic intracranial hemorrhage within 48 hours occurred in 1 patient (0.9%) in the tirofiban group and in no patients (0%) in the placebo group. The rate of any radiologic intracranial hemorrhage within 48 hours was 1.7% in the tirofiban group and 0% in the placebo group. The 90-day mortality was 0.6% in the tirofiban group and 1.6% in the placebo group (hazard ratio, 0.35; 95% CI, 0.04-3.33; P = .36) (Table 2). Other adverse events are presented in eTable 6 in Supplement 3.

Table 1. Baseline Participant Characteristics

Characteristics	Tirofiban (n = 177)	Placebo (n = 182)
Age, mean (SD), y	66 (12)	67 (10)
Sex, No. (%)	108 (61.0)	110 (60.4)
Female	69 (39.0)	72 (39.6)
Male	108 (61.0)	110 (60.4)
Score of 0 on modified Rankin Scale before stroke onset, No. (%)[a]	171 (96.6)	175 (96.2)
Pretenecteplase imaging, No. (%)		
Computed tomography	164 (92.7)	167 (91.8)
Magnetic resonance imaging	13 (7.3)	15 (8.2)
Medical history, No. (%)		
Hypertension	142 (80.2)	148 (81.3)
Diabetes	64 (36.2)	62 (34.1)
Hyperlipidemia	59 (33.3)	56 (30.8)
Ischemic stroke	32 (18.1)	24 (13.2)
Antiplatelet use	15 (8.5)	20 (11.0)
Transient ischemic attack	3 (1.7)	2 (1.1)
NIHSS score, median (IQR)[b]	6 (5-9)	6 (5-9)
ASPECTS, median (IQR)[c]	10 (9-10)	9 (9-10)
Qualifying insufficient response, No. (%)		
No neurological change	143 (80.8)	140 (76.9)
Neurological fluctuation	22 (12.4)	23 (12.6)
Neurological deterioration	12 (6.8)	19 (10.4)
Cause of stroke, No. (%)[d]		
Small artery occlusion	125 (70.6)	138 (75.8)
Large artery atherosclerosis	21 (11.9)	19 (10.4)
Cardioembolism	0	2 (1.1)
Unknown	31 (17.5)	23 (12.6)
Duration, median (IQR), min		
From last known well time to thrombolysis	180 (120-242)	180 (122-227)
From last known well time to randomization	638(460-1051)	623 (480-1001)
From last known well time to administration of tirofiban or placebo	660 (475-1086)	640 (496-1016)
From thrombolysis completion to randomization	392 (274-861)	448 (287-827)
From thrombolysis completion to administration of tirofiban or placebo	424 (290-874)	471 (314-835)
Oral antiplatelet therapy started 24-48 h after thrombolysis, No. (%)		
Monotherapy	96 (54.2)	105 (57.7)
Dual therapy	78 (44.1)	75 (41.2)
None	3 (1.7)	2 (1.1)

[a] Scores on the modified Rankin Scale of functional disability range from 0 (no symptoms) to 6 (death). A score of 1 indicates an ability to carry out all usual activities despite some symptoms. The modified Rankin Scale score before stroke onset was evaluated by site investigators using information obtained from patients (if possible) or their family members. Only patients with a prestroke modified Rankin Scale score of 0 to 1 were included in the study.

[b] Scores on the National Institutes of Health Stroke Scale (NIHSS) range from 0 to 42, with higher scores indicating greater neurological deficits.

[c] The Alberta Stroke Program Early Computed Tomography Score (ASPECTS) is an imaging measure of the extent of ischemic stroke. Scores range from 0 to 10, with higher scores indicating a smaller ischemic core.

[d] The cause of stroke was evaluated on the basis of medical history, clinical characteristics, and imaging results.

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Table 2. Efficacy and Safety Outcomes

Outcomes	Tirofiban (n = 177)	Placebo (n = 182)	Risk difference, % (95% CI)	Treatment effect (95% CI)[a]	*P* value
Primary efficacy outcome					
Modified Rankin Scale score of 0 to 1 at 90 d, No. (%)[b]	113 (63.8)	95 (52.2)	12 (1 to 22)	RR: 1.22 (1.02-1.46)	.03
Secondary clinical outcomes					
Modified Rankin Scale score at 90 d, No. of wins/total No. of pairs (%)[b]	14 200/32 214 (44.1)	10 281/32 214 (31.9)		Generalized OR: 1.38 (1.01-1.89)	.04
Modified Rankin Scale score of 0 to 2 at 90 d, No. (%)[b]	139 (78.5)	131 (72.0)	7 (−2 to 16)	RR: 1.09 (0.97-1.23)	.14
Modified Rankin Scale score of 0 to 3 at 90 d, No. (%)[b]	165 (93.2)	163 (89.6)	4 (−2 to 10)	RR: 1.04 (0.98-1.11)	.21
Early neurological improvement, No. (%)[c]	96 (54.2)	91 (50.0)	4 (−6 to 15)	RR: 1.08 (0.89-1.32)	.42
EQ-5D-5L score at 90 d, No. of wins/total No. of pairs (%)[d]	17 525/32 037 (54.7)	12 348/32 037 (38.5)		WR: 1.42 (1.08-1.83)	.03
Safety outcomes					
Symptomatic intracranial hemorrhage within 48 h, No./total (%)					
Heidelberg Bleeding Classification[e]	1/115 (0.9)	0/102	1 (−1 to 3)	RR: 2.66 (0.11-65.32)	.55
ECASS-2[f]	1/115 (0.9)	0/102	1 (−1 to 3)	RR: 2.66 (0.11-65.32)	.55
Any radiologic intracranial hemorrhage within 48 h, No./total No. (%)[e]	2/115 (1.7)	0/102	2 (−1 to 4)	RR: 4.44 (0.21-92.37)	.34
Mortality at 90 d, No. (%)	1 (0.6)	3 (1.6)		HR: 0.35 (0.04-3.33)	.36

Abbreviations: HR, hazard ratio; OR, odds ratio; RR, risk ratio; WR, win ratio.

[a] These unadjusted effect sizes compare the tirofiban group with the placebo group.

[b] Scores on the modified Rankin Scale of functional disability range from 0 (no symptoms) to 6 (death). Data were missing for 1 patient in the placebo group. The score was evaluated centrally by 2 modified Rankin Scale–certified neurologists who were blinded to treatment allocation and who reviewed the video or voice recordings elicited using a structured assessment.

[c] Defined as when the National Institutes of Health Stroke Scale (NIHSS) score at 48 hours after randomization was reduced by 30% or more compared with the NIHSS score at randomization.

[d] The European Quality of Life 5-Dimension 5-Level scale (EQ-5D-5L) is a continuous scale measuring self-reported quality of life. Scores range from −0.39 to 1; lower scores indicate worse quality of life. Data were missing for 1 patient in the placebo group.

[e] Intracranial hemorrhage was adjudicated by a clinical events committee. Symptomatic intracranial hemorrhage was assessed according to the Heidelberg Bleeding Classification criteria.

[f] This is a post hoc analysis in which symptomatic intracranial hemorrhage was defined according to the European Cooperative Acute Stroke Study (ECASS) 2 criteria.

Discussion

In patients with acute ischemic stroke, an insufficient early clinical response to intravenous tenecteplase, and no large or medium vessel occlusion or cardioembolic stroke, the INSTANT randomized clinical trial showed that adjunctive intravenous tirofiban led to a higher proportion of patients with excellent outcome at 90 days compared with placebo. Safety profiles were comparable between the 2 groups, with similarly low rates of symptomatic intracranial hemorrhage and mortality.

The beneficial treatment effects observed in this trial may be explained by the pathophysiological rationale for targeting platelet-mediated thrombosis during the vulnerable period following thrombolysis. Intravenous thrombolysis, while promoting clot dissolution, can paradoxically trigger platelet activation through 2 mechanisms: first, by reexposing the underlying atherosclerotic plaque during thrombus breakdown, and second, by releasing thrombin and highly activated platelets from the dissolving clot into the circulation. These processes could contribute to early vessel reocclusion and early neurological deterioration, particularly in patients showing an insufficient response to initial thrombolytic therapy.[6,20-22] Tirofiban, as a rapid-onset glycoprotein IIb/IIIa inhibitor, addresses this mechanism by selectively blocking the final common pathway of platelet aggregation, thereby potentially maintaining patency of the microvasculature during this critical period. In an open-label randomized clinical trial involving patients with acute noncardioembolic stroke presenting within 24 hours of symptom onset, intravenous tirofiban as a stand-alone therapy reduced the risk of early neurological deterioration compared with aspirin.[23] In the present trial, the point estimate was consistent with improved early neurological outcomes in the tirofiban group; however, this finding did not reach statistical significance and should therefore be interpreted with caution.

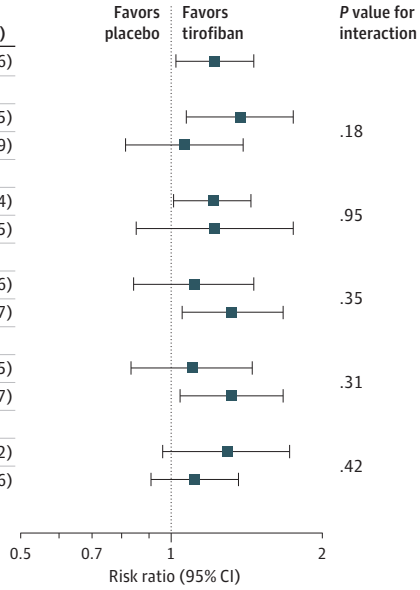
In addition, the patient selection criteria used in this trial likely contributed to the observed treatment effect. The RESCUE BT randomized, double-blind, placebo-controlled trial of patients with large vessel occlusion indicated no significant association between intravenous tirofiban administration prior to endovascular thrombectomy and reduced disability level.[24] Notably, subgroup analyses revealed a concerning safety signal, with tirofiban demonstrating a significant increase in symptomatic intracranial hemorrhage rates among patients with cardioembolic stroke.[25] By excluding individuals with large or medium vessel occlusion or cardioembolic stroke, the INSTANT trial focused on a population in whom platelet-rich microthrombosis plays a prominent pathophysiological role. This approach was based on the notion that

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Figure 2. Dot and Whisker Plot of Primary Outcome at 90 Days in Prespecified Subgroups

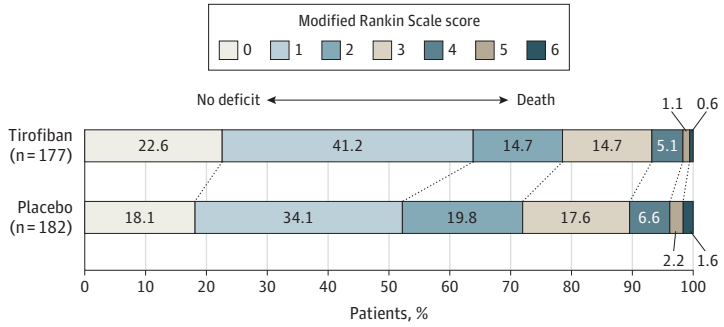
Subgroup	Modified Rankin Scale score of 0 to 1 at 90 d, No./total No. (%)		Risk difference, % (95% CI)	Risk ratio (95% CI)	*P* value for interaction
	Tirofiban	Placebo			
Overall	113/177 (63.8)	95/182 (52.2)	12 (1 to 22)	1.22 (1.02 to 1.46)	
Age, y					
≤67	67/97 (69.1)	45/89 (50.6)	19 (5 to 32)	1.37 (1.07 to 1.75)	.18
>67	46/80 (57.5)	50/93 (54.0)	4 (-11 to 18)	1.06 (0.81 to 1.39)	
Baseline NIHSS score					
≤6	75/93 (80.6)	62/93 (66.7)	14 (1 to 26)	1.21 (1.01 to 1.44)	.95
>6	38/84 (45.2)	33/89 (37.1)	8 (-6 to 23)	1.22 (0.85 to 1.75)	
Last known well time to randomization, min					
≤633	49/87 (56.3)	47/93 (50.5)	6 (-9 to 20)	1.11 (0.84 to 1.46)	.35
>633	64/90 (71.1)	48/89 (53.9)	17 (3 to 31)	1.32 (1.05 to 1.67)	
Last known well time to administration of tirofiban or placebo, min					
≤656	48/86 (55.8)	47/93 (50.5)	5 (-10 to 20)	1.10 (0.83 to 1.45)	.31
>656	65/91 (71.4)	48/89 (53.9)	18 (4 to 31)	1.32 (1.04 to 1.67)	
Oral antiplatelet therapy started 48 h after thrombolysis					
Monotherapy	52/96 (54.2)	44/105 (41.9)	12 (-2 to 26)	1.29 (0.96 to 1.72)	.42
Dual therapy	59/78 (75.6)	51/75 (68.0)	8 (-7 to 22)	1.11 (0.91 to 1.36)	



Age, baseline National Institutes of Health Stroke Scale (NIHSS) score, and durations of last known well time to randomization and to administration of tirofiban or placebo were dichotomized at the median. Scores on the NIHSS range from 0 to 42, with higher scores indicating worse neurological deficits.

Baseline NIHSS scores were obtained at the time of randomization. Unadjusted risk ratios were calculated and reported in accordance with the prespecified statistical analysis plan. Confidence intervals were not adjusted for multiple comparisons and cannot be used to infer treatment effects.

Figure 3. Bar Graph of Modified Rankin Scale Score Distribution at 90 Days in the Primary Analysis Population



Modified Rankin Scale scores range from 0 to 6, with 0 indicating no neurological deficit; 1, no clinically significant disability; 2, slight disability (able to handle own affairs without assistance but unable to carry out all previous activities); 3, moderate disability requiring some help (eg, with shopping, cleaning, and finances) but able to walk unassisted; 4, moderately severe disability (unable to attend to bodily needs without assistance and unable to walk unassisted); 5, severe disability (requiring constant nursing care and attention); and 6, death.

the composition of thrombi varies by stroke etiology, with noncardioembolic mechanisms being dependent on platelet-mediated thrombosis.[26] The difference in oral antiplatelet timing was inherent to the double-dummy design, essential for maintaining blinding. Active oral therapy was started at 24 hours in the placebo group but delayed until 44 hours in the tirofiban group to avoid confounding during the investigational infusion period. This design allows the comparison to attribute any early benefit specifically to tirofiban.

The risk of intracranial hemorrhage and mortality in this trial was very low. Only 1 patient experienced symptomatic hemorrhage, which had a fatal outcome. These results may be attributed to several factors. First, the trial predominantly enrolled patients with mild to moderate neurological deficits (median NIHSS score of 6) and small infarct volumes (Alberta Stroke Program Early Computed Tomography Score [ASPECTS] of 10), which inherently carries a lower risk of hemorrhagic transformation. Notably, the rate reported herein is substantially lower than the 3.2% symptomatic intracranial hemorrhage reported in the alteplase group of the PRISMS trial, which focused on minor stroke (NIHSS score of 0-5).[27] This discrepancy likely stems from key distinctions: INSTANT specifically enrolled patients who had already received tenecteplase and exhibited an inadequate clinical response—a

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different phenotype from the broader, treatment-naive minor stroke population in PRISMS. Second, the low-dose tirofiban regimen represented a conservative approach compared with previous studies. Also importantly, the protocol mandated a minimum 4-hour interval between tenecteplase completion and tirofiban initiation. Given tenecteplase's half-life of approximately 23 minutes, this delay ensured minimal pharmacodynamic overlap between the thrombolytic and antiplatelet agents, thereby reducing the risk of synergistic bleeding complications. Furthermore, any patient exhibiting neurological deterioration or fluctuation after tenecteplase was required to undergo urgent noncontrast computed tomography before randomization; this step excluded 21 patients with postthrombolysis hemorrhage from enrollment.

The results of this trial are consistent with those reported in the ASSET-IT trial.[28] A subgroup analysis of the RESCUE BT2 trial yielded similar results for patients who had intravenous thrombolysis followed by no neurological improvement.[29] However, INSTANT differs from these predecessors in several important aspects and provides complementary data. First, while ASSET-IT enrolled patients treated with either alteplase or tenecteplase, INSTANT specifically focused on those receiving tenecteplase, allowing a more homogeneous assessment of tirofiban in combination with this contemporary thrombolytic agent. Second, ASSET-IT included 293 patients (35.2%) with large or medium vessel occlusion, whereas INSTANT explicitly excluded such individuals. Third, unlike ASSET-IT, which administered tirofiban within 60 minutes after thrombolysis, INSTANT initiated treatment at a median of approximately 7.5 hours after thrombolysis, specifically targeting patients with an inadequate clinical response. Fourth, INSTANT used a lower tirofiban dose compared with the regimens used in both ASSET-IT and RESCUE BT2. To our knowledge, 2 ongoing trials (NCT06045156 and ChiCTR2400080653) are currently investigating similar therapeutic approaches, which may provide additional insights into optimal patient selection and treatment protocols. An individual participant data meta-analysis incorporating results from INSTANT, ASSET-IT, RESCUE BT2, and these ongoing studies may help clarify several key issues: first, the optimal timing of tirofiban initiation relative to thrombolysis; second, the most effective dosing strategy balancing efficacy and safety; third, identification of patient subgroups most likely to benefit from this therapeutic approach; and fourth, elucidation of the underlying mechanisms through which glycoprotein IIb/IIIa inhibition exerts its effects in the postthrombolysis period.

Limitations

This trial has several limitations. First, the interpretation of the treatment benefit must consider the precision of the estimate, as indicated by the confidence interval, and the observation that the significance was not maintained across all sensitivity analyses. This implies that although the point estimate suggests a clinically meaningful effect, the results still require confirmation through future larger-scale trials to validate their robustness and generalizability. Second, the exclusion of patients with minor stroke (NIHSS score <4), confirmed or suspected cardioembolic stroke, and large or medium vessel occlusion limit the generalizability of the findings. Third, 330 enrolled patients (91.9%) received intravenous tenecteplase within 4.5 hours of last known well time, leaving the efficacy and safety of early tirofiban added to thrombolytic treatment started in the extended time window unaddressed. Future studies are needed to explore whether similar benefits would be observed in patients treated beyond this time frame. Fourth, the interpretation of secondary outcomes is limited by the absence of multiplicity correction. The results for these end points should be interpreted as exploratory, with the primary outcome constituting the principal evidence of efficacy. Fifth, the exclusive enrollment of Chinese patients may restrict the applicability of the results in other populations known to differ in the frequency of intracranial atherosclerosis and other mechanisms.

Conclusions

This randomized clinical trial showed that among patients with acute noncardioembolic ischemic stroke and without large or medium vessel occlusion who had an inadequate clinical response to intravenous tenecteplase, adjunctive administration of intravenous tirofiban increased the likelihood of achieving excellent outcome at 90 days compared with placebo, without increasing the incidence of symptomatic intracranial hemorrhage or mortality. These findings suggest that early intravenous tirofiban may represent a potential adjunctive treatment following intravenous tenecteplase in patients with evidence of limited clinical response.

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